

NEONATAL MORTALITY, 2024-Q1

50.6 per 1,000 live births

50.6 per 1,000 is 2.7× the WHO / national benchmark of 19 per 1,000.

1,819 deaths across 35,916 live births, WHO GHO definition, 2 of 3 expected months, 2024-02 absent. 30 of 30 district figures are **PROVISIONAL**: 2024-01 and 2024-03 each loaded twice with differing values. Only 1 of 117 facilities stock surfactant, the standard treatment for prematurity; only 9 of 117 run an active quality-improvement programme; median self-reported reporting completeness is 76%. Capability gaps qualify whether this figure is actionable, not only whether it is settled.

Resolve the 2024-01 and 2024-03 batch conflict before treating this figure as final: DEFAULT-BATCH-01 is a placeholder default, not a decision.

MINISTRY OF HEALTH, RWANDA · NEONATAL · 2024-Q1

Quarterly Health Bulletin

2024-Q1

MONTHS
HELD

2/3

2024-02
absent

FACILITIES
REPORTING

117/117

in the months
held

FIGURES
PROVISIONAL

30/30

awaiting a batch
decision

REPORTING
COMPLETENESS,
MEDIAN

76%

range 50% to
99%, all facilities

Provisional figures are hatched in bars, hollow in dots, underlined in tables. Every check below always renders its finding: a caveat qualifies confidence, it never withholds content. Every figure links to its lineage record.

01

Facilities by delivery volume

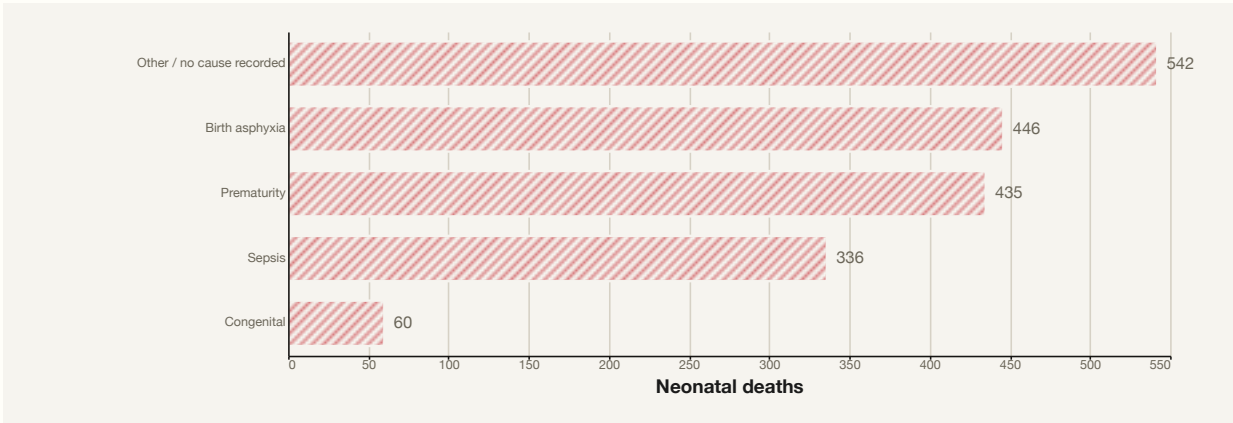


Top 10 of 117 by total deliveries. Volume tracks tier, so this is not a quality ranking. [lineage, section 8](#)

02

Neonatal deaths by cause, paired with the capability that treats it

The largest single bucket is **Other / no cause recorded**, 542 of 1,819 deaths (29.8%), ahead of every named clinical cause below it. That is a data-system finding, not a clinical one: it means the cause of death was not recorded, not that no cause exists. A reader taking the largest named clinical cause, Birth asphyxia, as the leading cause of neonatal death in this data would be wrong.



Ranked by count, not by assumed clinical severity. [lineage, section 8](#)

CAUSE	ICD-10	DEATHS	SHARE
Other / no cause recorded	P00-P96	542	29.8%

CAUSE	ICD-10	DEATHS	SHARE
Birth asphyxia Median competency 36% (range 10% to 96%) across 117 facilities	P21	446	24.5%
Prematurity Available at 1 of 117 facilities	P07	435	23.9%
Sepsis Usually or Always available at 14 of 117 facilities	P36	336	18.5%
Congenital Available at 10 of 117 facilities	Q00-Q99	60	3.3%

Reconciles to total neonatal deaths in 1,404 of 1,404 source rows. Capability figures are national, not quarter-scoped: capability data is a facility snapshot, section 3. [lineage, section 8](#)

Stillbirths, 2024-Q1: a different denominator, not a sixth cause row above. A stillbirth never had a live birth, so it cannot enter a live-birth-denominated rate and is not added to the neonatal-death count.

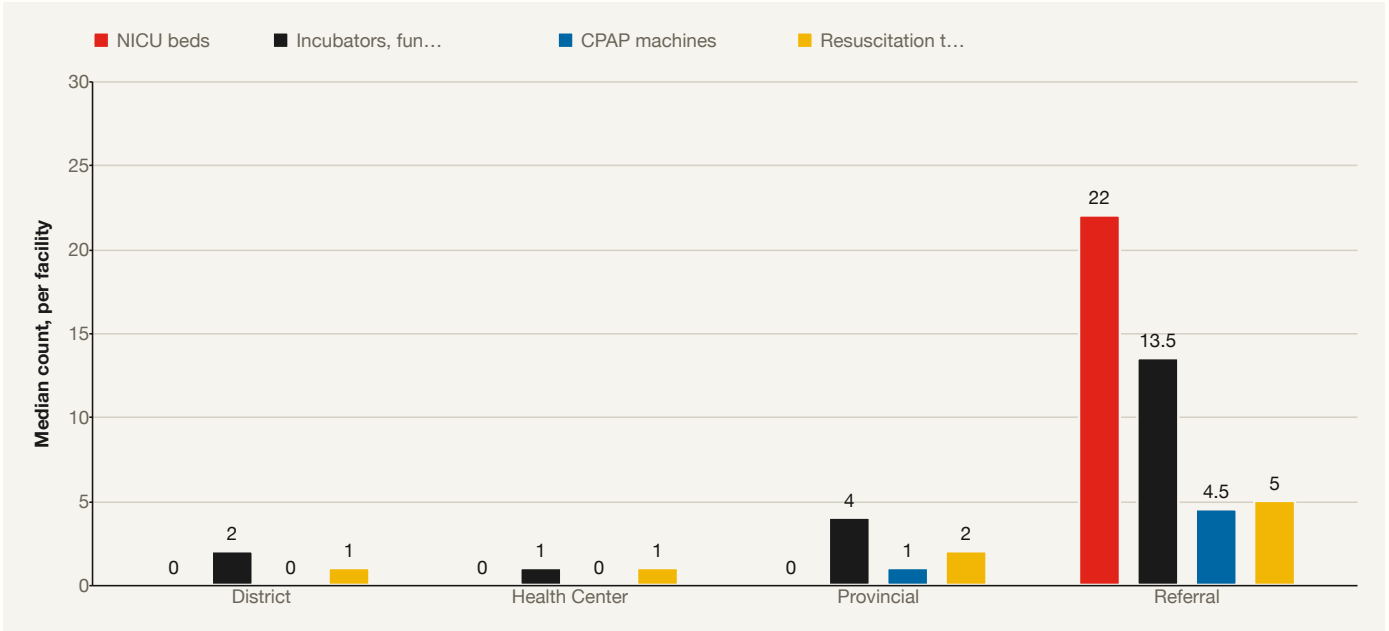
27.3

per 1,000 total deliveries · 1,008 of 36,924 · provisional

03

Facility capability, national, n = 117

Governance, operations and staffing data from all 117 facilities, resolved through the same [crosswalk](#) as every other figure on this page but never previously queried past the three covariates section 6 tests. Reporting completeness (median 76%) is the brief's own required "facility performance score." Broken out by tier below because tier, not any single equipment field, is what actually separates facilities in this data (section 6): District and Health Center read as near-identical on every metric in the table below, despite the widest mortality gap of any two adjacent tiers.



Median count per facility, by tier, NICU beds, Incubators, functional, CPAP machines, Resuscitation tables. [lineage, section 8](#)

Only 1 of 117 facilities stock surfactant and only 9 of 117 run an active quality-improvement programme, nationally; both are declared low-variance in section 7 and are not broken out by tier here for that reason. The wider capability set the brief asks for is below, by tier, in full:

METRIC	DISTRICT	HEALTH CENTER	PROVINCIAL	REFERRAL
n facilities	45	45	25	2
NICU beds (count)	0	0	0	22
Incubators, functional (count)	2	1	4	13.5
Incubators, total (count)	2	2	7	16.5
Radiant warmers (count)	2	2	4	9
Phototherapy units (count)	1	1	3	8.5
CPAP machines (count)	0	0	1	4.5
Resuscitation tables (count)	1	1	2	5
Oxygen cylinders available (count)	4	3	10	22
Oxygen concentrators (count)	1	1	3	11.5
Essential-drug stockout days (days/quarter)	23	22	14	2.5
Death audits conducted (%)	33	34	68	94.5
Supervision visits (/quarter)	1	1	2	3
Thermal-care protocol compliance (%)	24	22	47	78
Infection-prevention score (/100)	45	41	65	85.5
Nurses, total (count)	26	25	46	96
Nurses, neonatal-trained (count)	1	1	4	14
Midwives (count)	9	8	14	27.5
Obstetricians (count)	1	1	3	9.5
Pediatricians (count)	0	1	2	7.5
Anesthetists (count)	1	1	2	4.5

Median per facility, by tier, all facilities, all metrics the [crosswalk](#) resolves under this section that the chart above does not already carry. [lineage, section 8](#)

Neonatologists on staff: zero at all but 2 facilities, both Referral tier. The national mean is a rounding artefact of those two facilities, not staffing depth, section 7.

115/117

facilities with zero neonatologists

national, not quarter-scoped · [lineage, section 8](#)

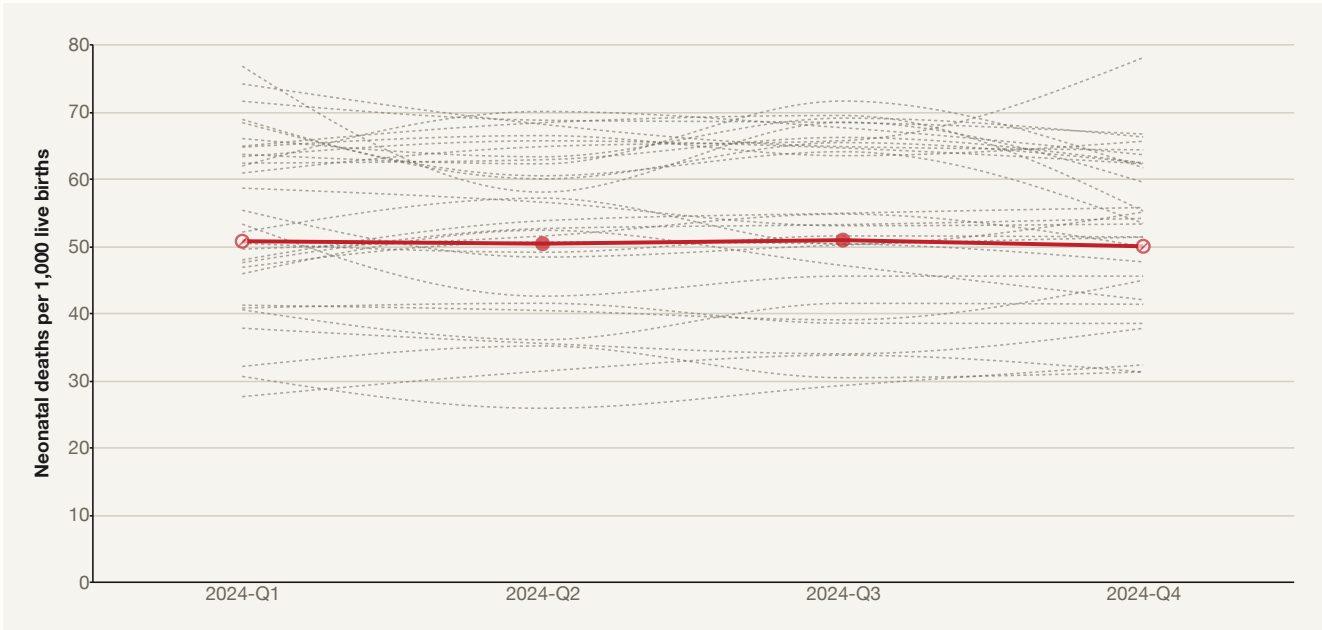
Neonatal mortality by district



30 districts, 27.5 to 76.7 per 1,000. District grain: facility coordinates are unusable, section 7. [lineage, section 8](#)

Trend, four quarters of 2024, by district

Not a two-point delta, and not national-only: every one of 30 districts, all four quarters, each labeled with the months actually held. The national series (bold) is the same four points this section showed before this edition; the district lines beneath it are new.



Thin lines: each district, dashed where that district-quarter figure is provisional (a batch conflict, section 7). Bold line: national, hatched points mark a partial quarter (a missing month, not a batch conflict). [lineage, section 8](#)

2024-Q1 (this edition) 50.6 per 1,000, national

1,819 deaths / 35,916 live births · 2 of 3 months held (partial quarter)

2024-Q2 50.3 per 1,000, national

2,734 deaths / 54,372 live births · 3 of 3 months held

2024-Q3 50.8 per 1,000, national

2,691 deaths / 52,965 live births · 3 of 3 months held

2024-Q4 49.9 per 1,000, national

1,792 deaths / 35,918 live births · 2 of 3 months held (partial quarter)

Caveat: observed lag-1 -0.091 does not exceed a within-facility permutation null of -0.107 ± 0.028 ; treat as directional, not conclusive. The chart above shows the real quarter-over-quarter series regardless: the caveat qualifies whether that movement is distinguishable from noise, it does not remove the series. Read the crossing, directionless district lines as that caveat made visible, not as evidence a trend is being hidden.

[lineage, section 8](#)

Tier confound: capability looks predictive, until you stratify

District facilities, nominally the mid-tier, have the worst mean neonatal mortality of any tier: 67.4 per 1,000 (n=45), ahead of Health Center (62.2, n=45), Provincial (32.5, n=25) and Referral (14.7, n=2).

A composite equipment index correlates -0.87 with facility mortality pooled across all 117 facilities, a relationship strong enough to read as "equip facilities, save newborns." It is not that. Stratified by tier it collapses to near zero or reverses (District -0.071; Health Center +0.036; Provincial -0.288): District and Health Center hold almost identical equipment (section 3) and still land 5.2 points apart in mortality. The pooled number describes which tier a facility is in, not what equipping it would do. Every covariate below is tested the same way, pooled and within tier: shown here, not hidden. Every correlation below also tests the same static, national capability snapshot (section 3) against each facility's births-weighted mortality rate summed across all four quarters of 2024, not matched quarter by quarter against that snapshot.

Referral closes a similar gap between two figures that sound alike. Whether a referral gets a reply pools at -0.81 against mortality; how fast it happens pools at +0.13. Neither survives stratification (below): the loop closing and the loop's speed both read as tier artifacts in this data, not real drivers of mortality once tier is held constant.

Staff trained on protocol (%) vs. neonatal mortality rate, stratified by tier



STRATUM	R
pooled, all facilities	-0.844
District	+0.111
Health Center	+0.042
Provincial	+0.144

Caveat: pooled $r=-0.844$ does not survive stratification by tier (District +0.111; Health Center +0.042; Provincial +0.144); the association describes tier, not the covariate

CPAP machines vs. neonatal mortality rate, stratified by tier



STRATUM	R
pooled, all facilities	-0.811
Provincial	-0.269

Caveat: pooled $r=-0.811$ does not survive stratification by tier (Provincial -0.269); the association describes tier, not the covariate

Average referral time (hrs) vs. neonatal mortality rate, stratified by tier



STRATUM	R
pooled, all facilities	+0.125
District	+0.046
Health Center	+0.306
Provincial	+0.273

Caveat: pooled $r=+0.125$ does not survive stratification by tier (District $+0.046$; Health Center $+0.306$; Provincial $+0.273$); the association describes tier, not the covariate

Composite equipment index (8 fields, z-scored) vs. neonatal mortality rate, stratified by tier



STRATUM	R
pooled, all facilities	-0.867
District	-0.071
Health Center	+0.036
Provincial	-0.288

Caveat: pooled $r=-0.867$ does not survive stratification by tier (District -0.071 ; Health Center $+0.036$; Provincial -0.288); the association describes tier, not the covariate

Referral feedback rate (%) vs. neonatal mortality rate, stratified by tier



STRATUM	R
pooled, all facilities	-0.809
District	-0.028
Health Center	+0.118
Provincial	-0.018

Caveat: pooled $r=-0.809$ does not survive stratification by tier (District -0.028 ; Health Center $+0.118$; Provincial -0.018); the association describes tier, not the covariate

[lineage, section 8](#)

07

Known issues in the source data

FINDING	DETAIL	EXTENT	HANDLING
gps_lat, gps_lon	Uniform random within Rwanda's bounding box. Every province spans the full country extent.	117 / 117	NOT USED

FINDING	DETAIL	EXTENT	HANDLING
2024-01, 2024-03	Each loaded twice with differing values. No timestamp distinguishes them, and both are internally consistent.	234 rows	QUEUED
facility_name	Contradicts tier_level .	62 / 117	LABEL ONLY
staff_per_delivery_2024	Named as derived but unreconstructable. Best of eight formulas matched 58.1%, worse than guessing the column's own mode.	3 values	NOT USED
capability_kangaroo_care_space, ops_kangaroo_practiced	Kangaroo-care self-contradiction: kangaroo_care_space contradicts kangaroo_care_practiced in 19 of 32 cases (59.4%).	n/a	CONTRADICTION
staff_nurses_neonatal, staff_last_training	Trained-nurses-vs-never-trained: 36 of 48 facilities reporting trained neonatal nurses have last_neonatal_training_date = 'Never'.	n/a	CONTRADICTION
ops_referrals_out, ops_referrals_in	Referral imbalance: Aggregate outbound referrals exceed inbound by 74% across a closed network.	n/a	CONTRADICTION
capability_electricity_reliable, capability_backup_generator	Power without generator: 33 of 117 facilities have neither reliable power nor a backup generator while holding powered equipment.	n/a	CONTRADICTION
staff_night_coverage	Night-shift coverage unreported: night_shift_coverage is blank for 63 of 117 facilities (53.8%). The column records only Full (9) and Partial (45); it has no value meaning 'none'. A blank is an unanswered question, not an absence of coverage, and is not counted as either.	n/a	MISSING DATA
ops_surfactant	Surfactant availability, no variance: surfactant_available is 'No' in 116 of 117 facilities. The column cannot separate facilities and no comparison built on it would mean anything.	n/a	LOW VARIANCE
staff_neonatologists	Neonatologist staffing, no variance: neonatologists is 0 in 115 of 117 facilities. The national mean is a rounding artefact of two facilities and should not be read as staffing depth.	n/a	LOW VARIANCE
governance_qi_active	Quality-improvement programme, no variance: quality_improvement_active is 'No' in 108 of 117 facilities. Present as a governance field, absent as a governance practice.	n/a	LOW VARIANCE
governance_*, ops_*, staff_*, capability_*	Snapshot tables joined to a monthly panel: facilities, governance, operations, and healthcare_workers are one row per facility with no date column; only clinical_neonatal is a monthly panel. Every governance_*, ops_*, staff_*, and capability_* figure in facility_capability and capability_summary is joined onto the 12-month clinical panel assuming constant-all-year state, an assumption the source data cannot verify. This is load-bearing for section 6's correlations, which match these assumed-constant covariates against a mortality rate that varies by quarter.	n/a	GRAIN
ops_referrals_out, ops_referrals_in	Named monthly, recorded without a period: referrals_out_monthly and referrals_in_monthly are named for a monthly grain, but operations.csv carries no date or period column at all. The suffix cannot be checked against the source and is assumed, not verified.	n/a	GRAIN

FINDING	DETAIL	EXTENT	HANDLING
governance_staff_trained_rate, capability_cpap_machines, ops_referral_time_hrs	Static capability against an annual rate: Section 6's correlations pair each covariate's static, all-year capability value against a neonatal mortality rate summed across every available period, not matched to any specific quarter. The result is a pooled, annual comparison, not a quarter-scoped one.	n/a	GRAIN
avg_gestational_age, preterm_births_28_32w, preterm_births_32_37w, apgar_less_7_at_5min, birth_weight_less_2500g	Excluded columns, reasons recorded: Five gestational and birth-outcome columns are excluded from every mart and never mentioned in the bulletin. avg_gestational_age carries no information (uniform-sd-ratio 1.004, independent of the preterm counts it should predict); preterm_births_28_32w and preterm_births_32_37w are independent of avg_gestational_age and of each other, drawn separately by the generator; apgar_less_7_at_5min is independent of every clinical covariate it should track; birth_weight_less_2500g correlates r=0.018 with prematurity where reality gives 0.6-0.9. Each is disqualified on evidence recorded in crosswalk.csv, not dropped silently.	n/a	EXCLUDED
governance_protocol_updated	Protocol date carried, never rendered: protocol_last_updated (canonical governance_protocol_updated) is carried into facility_capability and capability_summary but is never queried or rendered anywhere in the bulletin or email edition.	n/a	NOT RENDERED

Contradictions read live off `gold.known_contradictions`, sourced from the [crosswalk](#)'s own notes.

08 Lineage

Of 66 source columns in the [crosswalk](#), 52 become observations, 5 are identity or dimension keys, and 9 are excluded, each with a recorded reason. One observation, the raw protocol-update date, reaches a gold table and is never rendered directly: the crosswalk's own notes record it as redundant, since the protocol-status field this page does render is a pure function of that date's staleness (outdated at 24 months or more, no exceptions in 103 facilities checked).

FIGURE	GRAIN	INPUTS	RULE APPLIED	DEFINITION
NEONATAL MORTALITY, HEADLINE	national, quarter	35,916 live births, 1,819 deaths, 30 districts, 2 of 3 expected months, 2024-02 absent	DEFAULT-BATCH-01 lowest occurrence ordinal, arbitrary and pending resolution	WHO GHO: (early + late neonatal deaths) / live births × 1,000
FACILITIES BY DELIVERY VOLUME	facility, quarter	117 facilities, 2 months each, top 10 shown	DEFAULT-BATCH-01 on contested batches	Sum of <code>total_deliveries</code> over held months
NEONATAL DEATHS BY CAUSE	national, quarter	5 cause elements, reconciling in 1,404 of 1,404 source rows	DEFAULT-BATCH-01 on contested batches	ICD-10 P21, P07, P36, Q00 to Q99, summed over held months
STILLBIRTHS	national, quarter	1,008 stillbirths, 36,924 total deliveries	DEFAULT-BATCH-01 on contested batches	ICD-10 P95, per 1,000 total deliveries, WHO GHO denominator (not live births)
FACILITY CAPABILITY	national, all facilities	42 governance/operations/staffing/capability elements, period = ALL, 140 metric rows	no batch resolution, one snapshot per facility	<code>gold.facility_capability</code> : pivot of <code>observations_resolved</code> at period = ALL

FIGURE	GRAIN	INPUTS	RULE APPLIED	DEFINITION
NEONATAL MORTALITY BY DISTRICT	district, quarter	30 districts, 35,916 live births, 30 provisional	DEFAULT-BATCH-01 lowest occurrence ordinal	WHO GHO per district, benchmarked against 19 per 1,000
TREND, FOUR QUARTERS, BY DISTRICT	national + district, quarter, all 2024	9,036 deaths across all four quarters, 30 districts × 4 quarters	seed 20260810, 200 permutation trials; DEFAULT-BATCH-01 on contested district-quarter batches	checks.temporal_signal_check : lag-1 autocorrelation vs. within-facility permutation null
CORRELATION, MORTALITY DRIVERS	facility, all quarters pooled	5 covariates tested against tier	stratified by tier , survives only if sign and half-magnitude hold in every stratum with $n \geq 10$	checks.stratification_check (3) + web/src/lib/mart.ts equipment index and referral covariates (2, this layer, pipeline out of scope): pooled vs. within-tier Pearson r
MEAN MORTALITY BY TIER	facility, all quarters pooled, grouped by tier	117 facilities across 4 tiers	no batch resolution beyond nmr_facility_quarter 's own	mean of per-facility (deaths ÷ births × 1,000), grouped by org_units.tier

Generated from assignment_csv and dhis2 · 30 of 30 district figures provisional · every check above always renders its finding, with a caveat, never a gate

Pipeline run.py (Hamilton, DuckDB, Parquet) · site Astro, zero client JavaScript

Rebuild: uv run python run.py && bun run build